

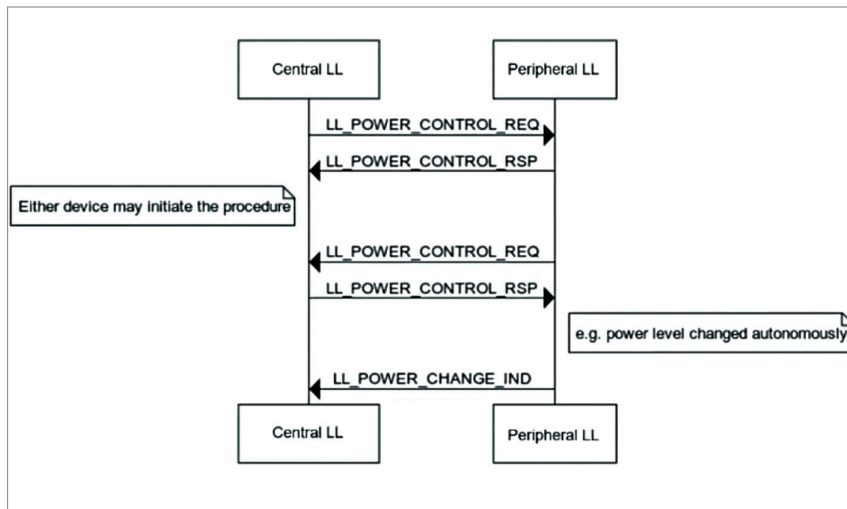


Figure 9: Flow control

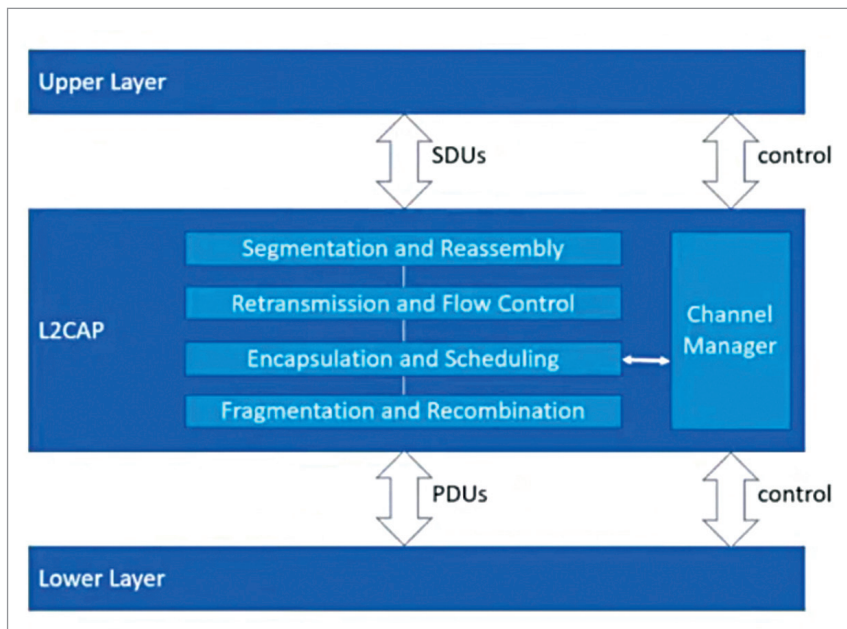


Figure 10: Flow control technique

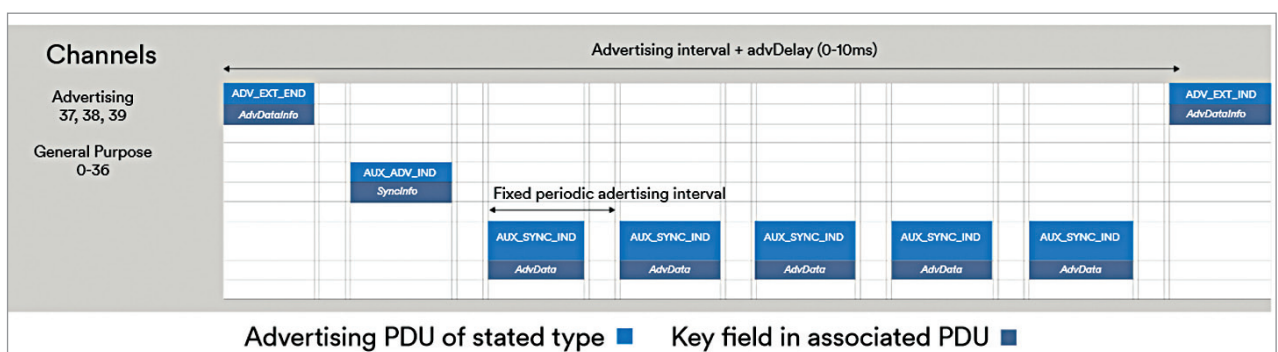


Figure 11: Periodic advertising

Reliable handling of Bluetooth communications

While connecting, devices agree on certain communication parameters, including several that are related to timing, such as how often a radio can be used for servicing this connection. The radio has to be shared potentially across lots of different connections. This particular parameter is called the connection interval.

At the start of the connection interval, devices take turns to transmit and receive with a gap of 150 microseconds in between packets. The precise, agreed timing parameters that control connections provide coordination between the transmit and receive behaviours of the two devices, which makes communication reliable.

Connection oriented communication

Spread spectrum in connection oriented communication looks like this: Bluetooth LE divides the 2.4GHz band into 40 channels, each 2MHz wide. Three of them are used for advertising (that's dedicated to connectionless communication). The other 37 are used when two connected devices communicate.

At each connection event, the used channel changes at random. So, in this way, packet transmission gets dotted around the ISM band. In addition, the system notes any non-performing channel (due to CRC failures or spectrum interference) and removes